

Payman Sharafian

Robotics & Mechatronics Engineer | Ph.D. Candidate | Machine Learning & Perception

Louisville, KY • **Work Authorization: 4-year EAD (Green Card pending), no sponsorship required** • paymansharafian.github.io

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PROFESSIONAL SUMMARY

Robotics and mechatronics engineer with 8 years of research and system-integration experience in physical human-robot interaction, teleoperation, safety-critical shared control, sensor fusion, and deep learning. Currently lead engineer on an NSF-funded mobile manipulator, responsible for integrating its motion control, safety filtering, embedded firmware, perception, and human-subject testing. Experienced in taking control and machine learning methods from theory to working hardware, and testing them with the people who use the robot.

TECHNICAL SKILLS

Languages: Python, C, C++ (C++14/17, STL), MATLAB, C#, TypeScript/JavaScript, Bash

Robotics & Autonomy: ROS1 (Kinetic/Melodic/Noetic), ROS2 (Humble), Gazebo, RViz, Motion & Path Planning, MoveIt, ROS Navigation Stack (AMCL, costmaps), TF2, Kinova Kortex, Mecanum omnidirectional kinematics, URDF, NVIDIA Isaac Sim

Control & Optimization: PID/PD Loop Tuning (classical control), MPC, Control Barrier Functions (CBF-QP, MPC-CBF), OSQP, CasADi, Admittance/Impedance Control, Neuroadaptive Control, Lyapunov Stability Analysis, Genetic Algorithms (PyGAD), Real-Time Control, Kalman Filtering

Perception & ML: PyTorch, OpenCV, TensorFlow, Keras, Scikit-learn, ONNX, CUDA, Semantic Segmentation (zero-shot, FastSAM), Temporal Convolutional Networks, BiLSTM, CNN-GRU, MLP, PointNet++, Contact-GraspNet, Reinforcement Learning, multi-modal tactile + force/torque fusion

Embedded & Hardware: Linux, real-time embedded firmware (RTOS), series elastic actuators (SEA), EtherCAT Protocol, CAN, SPI, I2C, UART, Arduino, Teensy, STM32 Nucleo, Sabertooth motor drivers, Jetson, VersaLogic blackbird board, analog signal conditioning, SolidWorks (DFM/assembly), FDM and SLA 3D printing, PCB design/debug/soldering, FSR tactile arrays (FlexiForce A201), ATI Axia 80 F/T sensors, multi-channel data acquisition (DAQ), RealSense RGB-D, 3D LiDAR

Software & Infrastructure: distributed system architecture, CI/CD (GitHub Actions), Next.js (App Router), React, TypeScript, Tailwind CSS, WebSockets, rosbridge, Cloudflare, CBOR/JPEG stream compression, socket programming, Docker, Git

PROFESSIONAL EXPERIENCE

Robotics Research Engineer (Ph.D. Research Assistant)

Sep. 2022 - Present

Louisville Automation & Robotics Research Institute (LARRI), University of Louisville

Louisville, KY

- Lead engineer on ARNA (Adaptive Robotic Nursing Assistant), an omnidirectional mobile manipulator funded by the NSF Future of Work program (#FW-HTF-RM) and trialed on an active hospital unit. ARNA pairs a 7-DOF Kinova Gen3 arm with a Mecanum base. Responsible for its control, perception, embedded firmware, web infrastructure, and clinical trials.
- Built ARNA's remote-teleoperation stack, covering network-aware safety filtering, secure browser access, and one-click autonomous grasping. Tested it with **30 remote operators** in several U.S. cities, with every command crossing the public internet.
 - Designed a five-layer shared-control stack that keeps hard safety guarantees in remote teleoperation even when the operator and the network connection are both unreliable. The adaptive layers can only make the robot more cautious and never relax the barrier constraints. The 100 Hz arm MPC-CBF had **zero boundary violations in 60 runs**, with jerk cost **200-440× lower** than raw operator input. The 50 Hz OSQP base filter, working on LiDAR sectors, had zero margin violations, and a watchdog widened the margins as the link degraded, with a 3.0 s dwell before recovery.
 - Wrote a TypeScript/Next.js teleoperation interface that operators open from a URL, with nothing to install. The robot has no inbound port open; access goes through an outbound Cloudflare Tunnel with identity-based Access. The interface compiles to a static bundle, so the robot runs no Node server. Split each session across three separate rosbridge instances so camera bursts cannot cause head-of-line blocking of joystick or stop commands, and cut per-frame data volume by **~35-40×** with JPEG encoding at the source and CBOR subscriptions.
 - Designed a semi-autonomous pick pipeline in which the remote operator only clicks the target object. FastSAM segments the object on CPU, and Contact-GraspNet with a PointNet++ encoder proposes ranked SE(3) grasps. Fit the pipeline onto a **4 GB laptop GPU (GTX 1060)** by cropping inference to the object's bounding region ($\sim 10\times$ less input) and loading the weights once at node start. Ranked grasps by reachability rather than network confidence, since reachability was what actually limited success.
- Led a sensorized handlebar and real-time adverse-event detector through mechanical design, electronics, ML, and human trials. The handlebar uses **eight FlexiForce A201 FSRs** and a custom Arduino signal-conditioning board publishing at **122 Hz**, fused with 6-axis F/T data through a Singer-model Kalman filter. Compared TCN, BiLSTM, and CNN-GRU models under subject-wise cross-validation, then deployed the TCN to ONNX on the robot itself (no cloud), where it re-evaluates every **8.2 ms**. On **10 unseen participants** it detected **85.0%** of events at **2.05 s mean latency**.
- Designed PNNUI, a parallel neural network teleoperation interface that maps any m -DOF input device onto any n -DOF robot without a kinematic model of either. It runs two networks side by side: a genetic algorithm tunes the first offline to minimize task time, and the second trains online by SGD to reduce measured jerk. In a 20-subject IRB study, **linear jerk halved** and **angular jerk fell about 3×** ($p < .001$), **with no increase in task-completion time**.

- Designed a model-free neuroadaptive admittance controller that learns ARNA's unmodeled dynamics online. The update laws come from a Lyapunov analysis, and the controller runs at 333 Hz against a 125 Hz admittance model. Compared with a tuned PD baseline in tests with 10 users, it improved tracking accuracy by up to **32%** and cut motion jerk by up to **52%**. Re-tuning it with **63 nursing students** brought velocity tracking error from **0.0157 to 0.0105 m/s** and mean actuator torque from **3.66 to 1.79 Nm**.
- Designed and led an acceptability trial with **10 patients and 5 nurses** on an active unit at University of Louisville Hospital. Combined logged task times with Technology Acceptance Model surveys and analyzed them with linear mixed-effects models in R. Participants rated the joystick significantly easier to use and more likely to be adopted than the tablet ($p \approx 0.02$ - 0.03), even though task times were statistically similar.
- Repaired and upgraded ARNA's motor-controller PCBs, brought non-functional sonar and bump sensors back online and into ROS, repurposed the LiDAR for navigation, and rebuilt onboard wiring across the EtherCAT motor chain, Sabertooth motor drivers, Teensy microcontrollers, battery/inverter bank, and instrumentation board. Worked across the robot's serial and fieldbus layers (EtherCAT, CAN, SPI, I2C, and UART) to bring sensors and drives back under ROS control. Maintained the distributed onboard-PC/Jetson software system.
- Maintained and extended ARNA's embedded layer, a custom motor-controller PCB with **two Teensy microcontrollers** that must stay synchronized so the Sabertooth motor drivers get coherent wheel commands. The board talks EtherCAT to the robot's main PC, and a separate STM32 Nucleo handles the safety sensors. Debugged and repaired this firmware across several hardware revisions.
- Mentored **seven undergraduate and two graduate researchers**, and wrote the lab manuals and Standard Operating Procedures for ARNA that new researchers use to get started.

Mechatronics Engineer (M.Sc. Research Assistant)

Nov. 2017 - Sep. 2021

Human-Robot Interaction Lab, University of Tehran

Tehran, Iran

- Redesigned and built the mechanical powertrain and embedded electronics of a 1-DOF upper-limb rehabilitation robot driven by a series elastic actuator (equivalent torsional stiffness **8-30 Nm/rad**). Replaced the belt drives with **08B-13** chains to stop slip under high load, added hardware safety interlocks (an ergonomic handle microswitch and an emergency stop), and integrated an Arduino Mega with **43 A** peak BTS7960 H-bridge drivers.
- Developed a torque-based PD impedance controller with active resistance that removed torque spikes at transitions and tracked trajectories within **6.16-8.68% mean steady-state torque error** across **8-16 Nm** resisting loads. Built a C#/.NET serial bridge that maps joint motion into interactive web and Python games, and validated the system in a **15-subject** user study with **>90% motivation** and **86% perceived safety** ratings.
- Wrote a sound-source localization algorithm for an interactive social robot used in autism therapy. Audio filtering improved **70%** and gesture detection accuracy rose **40%**.

EDUCATION

Ph.D. Candidate, Electrical & Robotics Engineering, University of Louisville

Sep. 2022 - **Dec. 2026 (expected)**

M.S. in Mechatronics Engineering, University of Tehran

Nov. 2017 - Sep. 2021

B.Sc. in Electrical Power Engineering, Islamic Azad University

Sep. 2012 - Aug. 2017

SELECTED PUBLICATIONS

Peer-Reviewed Journals

- Adaptive User Interface With Parallel Neural Networks for Robot Teleoperation. **P. Sharafianardakani**, M. A. Hanafy, I. Kondaurova, A. Ashary, M. M. Rayguru, D. O. Popa. *IEEE Robotics and Automation Letters*, 10(2):963-970, 2025.
- Evaluation of a Neuroadaptive Admittance Controller for Ambulation through Physical Human-Robot Interaction. **P. Sharafianardakani**, S. Abubakar, S. K. Das, S. Surupa, M. A. Hanafy, M. Rayguru, D. O. Popa. *Intelligent Service Robotics*, 18(7):1355-1368, 2025.
- Acceptance of an Adaptive Robotic Nursing Assistant for Ambulation Tasks. I. Kondaurova, **P. Sharafian**, R. Mitra, M. M. Rayguru, *et al.*, D. O. Popa. *Robotics (MDPI)*, 14(9):121, 2025.
- Perceived Usefulness of Robotic Technology for Patient Fall Prevention. M. C. Logsdon, I. Kondaurova, N. Zhang, *et al.* (incl. **P. SharafianArdakani**), D. O. Popa. *Workplace Health & Safety*, 72(12):542-549, 2024.

Under Review / In Preparation

- Tactile Handlebar and Deep Learning for Safe Interaction with a Robot Nursing Assistant. **P. Sharafianardakani et al.** *IEEE Transactions on Medical Robotics and Bionics* (**under review**).

Peer-Reviewed Conferences

- Hospital Trial Results for the Acceptability of an Adaptive Robot Nursing Assistant. **P. SharafianArdakani**, I. Kondaurova, A. Ashary, N. Zhang, M. C. Logsdon, M. Walker, D. O. Popa. *IEEE CASE*, pp. 2043-2049, 2025.

LEADERSHIP & AWARDS

Journal Reviewer, IEEE Robotics and Automation Letters (RA-L), 2024, 2025

Conference Reviewer, ACM/SIGAPP Symposium on Applied Computing (SAC), 2024

STEM Mentor, 7 undergraduate students and 2 graduate students, 2023-Present

Dissertation Completion Award (full tuition + stipend), University of Louisville, 2026

IEEE RAS Travel Support Award (ICRA 2025), IEEE Robotics and Automation Society, May 2025

Graduate Student Travel Award, University of Louisville, 2025